Advanced Object Oriented Programming

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高级面向对象编程

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变量 & 控制台 IO

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01

Variables

01

变量

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Variables

- Numbers that are not specified or may change in the future
- The most common way to **use memory**
- Causes the content of operations to vary depending on the values determined specifically
  - ex) Weather Variables
  - The Beginning of **Dependency**

变量

- 未指定或在将来可能变化的数字
- 使用内存的最常见方式
- 导致操作内容根据特定确定的值而变化
  - ex) 天气变量
  - 依赖的开始

Constants

- Numbers whose values are fixed and cannot be changed in the future
  - Literal Constants
    - ‘A’, “hello”, 3, 3.4F, 3.4
  - Symbolic Constants
    - final

常量

- 值在将来固定且不可更改的数字
  未来
  - 文字常量
    - ‘A’ , “hello” , 3, 3.4F, 3.4
  - 符号常量
    - final

Identifier

- Another way to describe a variable is "identifier."

- Consists of uppercase and lowercase English letters, “_", and numbers.
- The first character cannot be a number.
- Spaces are not allowed in the name.

标识符

- 变量的另一种描述方式是“标识符。”

- 由大写和小写英文字母、“_”，以及数字组成。

- 第一个字符不能是数字。
- 名称中不允许有空格。

Identifier

- Another way to describe a variable is "identifier."

- Consists of uppercase and lowercase English letters, “_", and numbers.
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标识符

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- 第一个字符不能是数字。
- 名称中不允许有空格。

Identifier

- Reserved words are not allowed.
  - int, for
- Excessively long names are prohibited (recommended).
  - Int multibytestringtowidesting
  - **Camel case** recommended
- Names with unclear meaning.
  - Int aaa, float bbb, char ccc
  - Factors that make engineers difficult

标识符

- 不允许使用保留字。
  - int , for
- 名称过长是不允许的（推荐）。
  - Int 多字节字符串转宽字符串
  - **推荐使用驼峰命名法**
- 含义不明确的名称。
  - Int aaa , float bbb , char ccc
  - 让工程师变得困难的因素

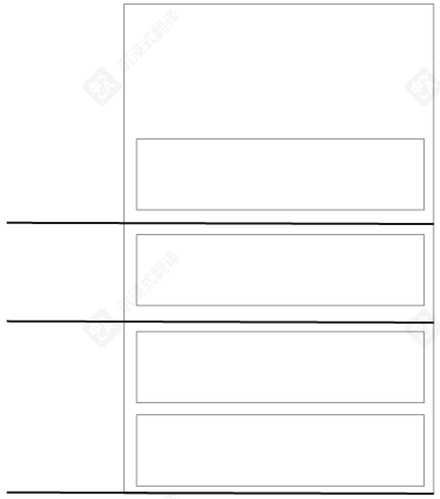
Types of Variables

- Local Variables
  - Stack Variables If Not Declared as Static
- Parameters
- Member Variables
- Class Variables

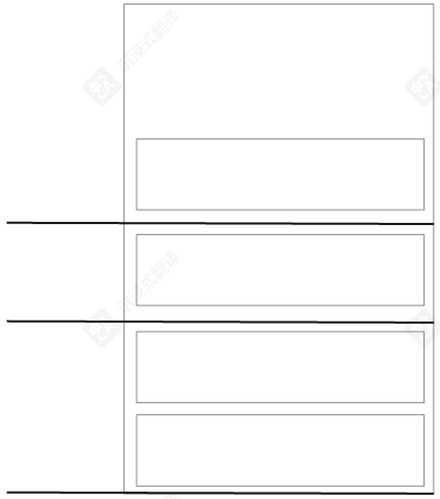
变量的类型

- 局部变量
  - 如果未声明为静态的栈变量
- 参数
- 成员变量
- 类变量

Local Variable with Stack 局部变量与栈



```
{
    int a;
    int b;
    {
        int c;
        {
            int d;
        }
    }
}
```



```
{
    int a;
    int b;
    {
        int c;
        {
            int d;
        }
    }
}
```

Code and Comments

- Grammar for leaving comments in programs
  - Maximizing **maintainability**

- Not included in program code
  - // Entire line
  - /* */ Entire section

代码和注释

- 在程序中添加注释的语法
  - **最大化可维护性**

- 未包含在程序代码中
  - // 整行
  - /* */ 整段

Quiz

- What are the names used to distinguish variables, classes, methods, etc. in programming?
  A. Literal
  B. Keyword
  C. Identifier
  D. Data Type

Quiz

- 用于区分编程中变量、类、方法等的名称是什么？等。
  A. 字面意思
  B. 关键字
  C. 标识符
  D. 数据类型

Quiz

- Which of the following is not a valid variable identifier?
  A. total_count
  B. _userName
  C. 7days
  D. myVariable

Quiz

- 以下哪个不是有效的变量标识符?
  A. total _count
  B. _userName
  C. 7days
  D. myVariable

Quiz

- What is the most important reason for naming variables and methods meaningfully?
  A. Increased program execution speed
  B. Improved code readability and maintainability
  C. Reduced memory usage
  D. Reduced security vulnerabilities

Quiz

- 为什么变量和方法要有意义地命名?

  A. 提高程序执行速度
  B. 提高代码可读性和可维护性
  C. 减少内存使用
  D. 减少安全漏洞

Quiz

- What are the two main memory areas used by the JVM, as explained in the lecture?
  A. CPU and GPU
  B. Cache and Register
  C. Stack and Heap
  D. RAM and ROM

Quiz

- JVM中使用的两个主要内存区域是什么，正如讲座中解释的那样？

  A. CPU和GPU
  B. 缓存和寄存器
  C. 栈和堆
  D. RAM和ROM

Quiz

- What is the JVM memory area that follows the LIFO (Last In, First Out) principle, stores local variables and other data when a function is called, and automatically manages memory?
  A. Heap
  B. Static
  C. Register
  D. Stack

Quiz

- 遵循后进先出（LIFO）原则的JVM内存区域是什么？它在函数调用时存储局部变量和其他数据，并自动管理内存？

  A. 堆
  B. 静态
  C. 注册
  D. 栈

02

Console

02

控制台

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Console and CLI

- CLI(Command Line Interface) based HCI(Human Computer Interface) is implemented using keyboard input.
- Keyboard inputs are stored sequentially in memory (**I/O buffer**).
- The I/O buffer processes characters (or keycodes) one by one.

控制台和CLI

- 基于CLI（命令行界面）的人机界面HCI是通过键盘输入实现的。
- 键盘输入按顺序存储在内存（I/O缓冲区）中。
- I/O缓冲区逐个处理字符（或键码）。

Console

控制台

A:\>dir
Volume in drive A has no label
Directory of A:\
COMMAND COM 25307 3-17-87 12:00p
ANSI SYS 1670 3-17-87 12:00p
FDISK COM 48216 3-18-87 12:00p
FORMAT COM 11616 3-18-87 12:00p
MODE COM 15487 3-17-87 12:00p
DISKCOPY COM 6525 3-17-87 12:00p
EDLIN COM 7526 3-17-87 12:00p
Q EXE 47936 7-27-11 1:36p
QCONFIG EXE 27779 11-18-83 2:06a
AUTOEXEC BAT 98 7-27-11 1:36p
10 File(s) 110592 bytes free
A:\>

A:\>dir
Volume in drive A has no label
Directory of A:\
COMMAND COM 25307 3-17-87 12:00p
ANSI SYS 1670 3-17-87 12:00p
FDISK COM 48216 3-18-87 12:00p
FORMAT COM 11616 3-18-87 12:00p
MODE COM 15487 3-17-87 12:00p
DISKCOPY COM 6525 3-17-87 12:00p
EDLIN COM 7526 3-17-87 12:00p
Q EXE 47936 7-27-11 1:36p
QCONFIG EXE 27779 11-18-83 2:06a
AUTOEXEC BAT 98 7-27-11 1:36p
10 File(s) 110592 bytes free
A:\>

Console and I/O

- Keycode values mostly correspond to **ASCII codes**.
- Read 1 byte at a time and return the value.
- Non-English languages use 2 bytes.

```
System.in.read()
```

控制台和I/O

- 键码值主要对应ASCII码值。
- 每次读取1个字节并返回该值。
- 非英语语言使用2个字节。

```
System.in.read()
```

Console and I/O

控制台和I/O

```java
public class Main {
    public static void main(String[] args) throws Exception {
        int keyCode = System.in.read();
        System.out.println(keyCode);
        keyCode = System.in.read();
        System.out.println(keyCode);
        keyCode = System.in.read();
        System.out.println(keyCode);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
ABC
65
66
67
```


```java
public class Main {
    public static void main(String[] args) throws Exception {
        int keyCode = System.in.read();
        System.out.println(keyCode);
        keyCode = System.in.read();
        System.out.println(keyCode);
        keyCode = System.in.read();
        System.out.println(keyCode);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
ABC
65
66
67
```

Console and I/O

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
가
234
176
128
```

- Non-English characters use 2 bytes.
- UTF-8 encoding results.
  - UTF-8 encodes Unicode values in 1-4 bytes.
- In Java, strings do not end with NULL.

控制台和I/O

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
가
234
176
128
```

- 非英文字符使用2个字节。
- UTF-8编码结果。
  - UTF-8以1-4个字节编码Unicode值。
- 在Java中，字符串不以NULL结尾。

Console and I/O

- How to check UTF-8 string

```java
public class Main {
    public static void main(String[] args) throws Exception{
        String ga = "가";
        System.out.println((int)'가');
        System.out.printf("%04X\n", (int)'가');
        byte[] codeData = ga.getBytes();
        System.out.printf("%02X %02X %02X\n",
                codeData[0], codeData[1], codeData[2]);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
44032
AC00
EA B0 80
```

控制台和I/O

- 如何检查UTF-8字符串

```java
public class Main {
    public static void main(String[] args) throws Exception{
        String ga = "가";
        System.out.println((int)'가');
        System.out.printf("%04X\n", (int)'가');
        byte[] codeData = ga.getBytes();
        System.out.printf("%02X %02X %02X\n",
                codeData[0], codeData[1], codeData[2]);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
44032
AC00
EA B0 80
```

Scanner class and Input buffer

扫描器类和输入缓冲区

s.nextInt()

s.nextDouble()

s.nextInt()

s.nextDouble()

Integer and real number input/output

整数和实数输入/输出

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int a = s.nextInt();
        double b = s.nextDouble();
        System.out.println("a: " + a);
        System.out.println("b: " + b);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
3 3.4
a: 3
b: 3.4
```


```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int a = s.nextInt();
        double b = s.nextDouble();
        System.out.println("a: " + a);
        System.out.println("b: " + b);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
3 3.4
a: 3
b: 3.4
```

Input Error

输入错误

```
실행  Main ×
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe "-javaagent:C:\Program F
3 a
Exception in thread "main" java.util.InputMismatchException Create breakpoint
    at java.base/java.util.Scanner.throwFor(Scanner.java:947)
    at java.base/java.util.Scanner.next(Scanner.java:1602)
    at java.base/java.util.Scanner.nextDouble(Scanner.java:2573)
    at Main.main(Main.java:7)

종료 코드 1(으)로 완료된 프로세스
```


```
실행  Main ×
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe "-javaagent:C:\Program F
3 a
Exception in thread "main" java.util.InputMismatchException Create breakpoint
    at java.base/java.util.Scanner.throwFor(Scanner.java:947)
    at java.base/java.util.Scanner.next(Scanner.java:1602)
    at java.base/java.util.Scanner.nextDouble(Scanner.java:2573)
    at Main.main(Main.java:7)

종료 코드 1(으)로 완료된 프로세스
```

Input Error

输入错误

```java
import java.util.Scanner;

//TIP To <b>Run</b> code, press <shortcut actionId="Run"/> or
// click the <icon src="AllIcons.Actions.Execute"/> icon in the gutter.
public class Main {
    public static void main(String[] args) {
        //TIP Press <shortcut actionId="ShowIntentionActions"/> with your caret at the highlighted text
        // to see how IntelliJ IDEA suggests fixing it.
        Scanner s = new Scanner(System.in);
        try{
            int a = s.nextInt();
            double b = s.nextDouble();
        }
        catch (Exception e){
            System.out.println("Error!");

        }

        System.out.println("End");
    }

}
```


```java
import java.util.Scanner;

//TIP To <b>Run</b> code, press <shortcut actionId="Run"/> or
// click the <icon src="AllIcons.Actions.Execute"/> icon in the gutter.
public class Main {
    public static void main(String[] args) {
        //TIP Press <shortcut actionId="ShowIntentionActions"/> with your caret at the highlighted text
        // to see how IntelliJ IDEA suggests fixing it.
        Scanner s = new Scanner(System.in);
        try{
            int a = s.nextInt();
            double b = s.nextDouble();
        }
        catch (Exception e){
            System.out.println("Error!");

        }

        System.out.println("End");
    }

}
```

Scanner class and Input buffer

扫描器类和输入缓冲区

s.nextInt()

s.nextLine()

1	0		t	e	s	t	\n		

s.nextInt()

s.nextLine()

1	0		t	e	s	t	\n		

Character and String number input/output 字符和字符串数字输入/输出

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) throws Exception{
        char ch = (char)System.in.read();
        Scanner s = new Scanner(System.in);
        String data = s.nextLine();

        System.out.println("ch: " + ch);
        System.out.println("data: " + data);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
a test
ch: a
data:  test
```


```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) throws Exception{
        char ch = (char)System.in.read();
        Scanner s = new Scanner(System.in);
        String data = s.nextLine();

        System.out.println("ch: " + ch);
        System.out.println("data: " + data);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
a test
ch: a
data:  test
```

Escape Sequence 转义序列

Escape Sequences

Escape Sequence	Description
\t	Insert a tab in the text at this point.
\b	Insert a backspace in the text at this point.
\n	Insert a newline in the text at this point.
\r	Insert a carriage return in the text at this point.
\f	Insert a formfeed in the text at this point.
\'	Insert a single quote character in the text at this point.
\"	Insert a double quote character in the text at this point.
\\	Insert a backslash character in the text at this point.

Escape Sequences

Escape Sequence	Description
\t	Insert a tab in the text at this point.
\b	Insert a backspace in the text at this point.
\n	Insert a newline in the text at this point.
\r	Insert a carriage return in the text at this point.
\f	Insert a formfeed in the text at this point.
\'	Insert a single quote character in the text at this point.
\"	Insert a double quote character in the text at this point.
\\	Insert a backslash character in the text at this point.

Escape Sequence 转义序列

```java
public class Main {
    public static void main(String[] args) {
        System.out.printf("A\n");
        System.out.printf("ABC\b\n");
        System.out.printf("\tABC\n");
        System.out.printf("\"Hello\"\n");
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
A
AB
    ABC
"Hello"
```


```java
public class Main {
    public static void main(String[] args) {
        System.out.printf("A\n");
        System.out.printf("ABC\b\n");
        System.out.printf("\tABC\n");
        System.out.printf("\"Hello\"\n");
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
A
AB
    ABC
"Hello"
```

Format Characters

Conversion Character	Displays Argument (Variable's Contents) As
%c	Single character
%d	Signed decimal integer (int)
%e	Signed floating-point value in E notation
%f	Signed floating-point value (float)
%g	Signed value in %e or %f format, whichever is shorter
%i	Signed decimal integer (int)
%o	Unsigned octal (base 8) integer (int)
%s	String of text
%u	Unsigned decimal integer (int)
%x	Unsigned hexadecimal (base 16) integer (int)

https://javasyntax.wordpress.com/2009/08/15/6-introduction-to-java-programs/

格式化字符

Conversion Character	Displays Argument (Variable's Contents) As
%c	Single character
%d	Signed decimal integer (int)
%e	Signed floating-point value in E notation
%f	Signed floating-point value (float)
%g	Signed value in %e or %f format, whichever is shorter
%i	Signed decimal integer (int)
%o	Unsigned octal (base 8) integer (int)
%s	String of text
%u	Unsigned decimal integer (int)
%x	Unsigned hexadecimal (base 16) integer (int)

https://javasyntax.wordpress.com/2009/08/15/6-java程序简介/

Format Characters

```java
public class Main {
    public static void main(String[] args) {
        System.out.printf("%c\n", 65);
        System.out.printf("%d\n", 1234);
        System.out.printf("+%d, %+d\n", 1234, -1234);
        System.out.printf("%8d\n", 1234);
        System.out.printf("%08d\n", 1234);
        System.out.printf("%s\n", "Hello");
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
A
1234
+1234, -1234
    1234
00001234
Hello
```

格式化字符

```java
public class Main {
    public static void main(String[] args) {
        System.out.printf("%c\n", 65);
        System.out.printf("%d\n", 1234);
        System.out.printf("+%d, %+d\n", 1234, -1234);
        System.out.printf("%8d\n", 1234);
        System.out.printf("%08d\n", 1234);
        System.out.printf("%s\n", "Hello");
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
A
1234
+1234, -1234
    1234
00001234
Hello
```

Format Characters

格式化字符

```java
public class Main {
    public static void main(String[] args) {
        double data = 123.345;
        System.out.printf("%f, %f\n", data, -data);
        System.out.printf("%.1f, %.2f\n", data, data);
        System.out.printf("%.3f, %.4f\n", data, data);
        System.out.printf("%012.2f\n", data);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
123.345000, -123.345000
123.3, 123.35
123.345, 123.3450
000000123.35
```


```java
public class Main {
    public static void main(String[] args) {
        double data = 123.345;
        System.out.printf("%f, %f\n", data, -data);
        System.out.printf("%.1f, %.2f\n", data, data);
        System.out.printf("%.3f, %.4f\n", data, data);
        System.out.printf("%012.2f\n", data);
    }
}
```

```
C:\Users\cx853\.jdks\openjdk-20.0.1\bin\java.exe
123.345000, -123.345000
123.3, 123.35
123.345, 123.3450
000000123.35
```

Quiz

- What is the basic interaction method for console input/output?
  A. Mouse clicks via a graphical interface
  B. Text-based command entry and results review
  C. Network packet transmission and reception
  D. Executing database queries

Quiz

- 控制台输入/输出的基本交互方式是什么?
  A. 通过图形界面进行鼠标点击
  B. 基于文本的命令输入和结果查看
  C. 网络数据包的发送和接收
  D. 执行数据库查询

Quiz

- What is the primary role of a "buffer" used in the input/output process?
  A. A security function that encrypts input data.
  B. A storage function that permanently stores data.
  C. A temporary storage space that mitigates the speed difference between input/output devices and the computer.
  D. A function that compresses and transmits output data.

Quiz

- “缓冲器”在输入/输出过程中的主要作用是什么?
  A. 一种加密输入数据的安防功能。
  B. 一种永久存储数据的存储功能。
  C. 一种临时存储空间，用于缓解输入/输出设备与计算机之间的速度差异。
  D. 一种压缩和传输输出数据的功能。

Quiz

- In Java, what input stream is most commonly used when reading formatted input, such as integers or strings, from the console using the `Scanner` class?
  A. System.out
  B. System.error
  C. System.in
  D. FileReader

Quiz

- 在 Java 中，当使用 `Scanner` 类从控制台读取格式化输入（如整数或字符串）时，最常用的输入流是什么？

  A. System.out
  B. System.error
  C. System.in
  D. FileReader

Quiz

- What is the syntax used to represent special characters like line breaks and tabs within string literals in Java?
  A. Format Specifiers
  B. Wildcard Characters
  C. Escape Sequences
  D. Regular Expressions

Quiz

- Java中用于表示字符串字面量中换行符和制表符等特殊字符的语法是什么?
  A. 格式说明符
  B. 通配符字符
  C. 转义序列
  D. 正则表达式

Quiz

- What are some common problems that can occur when calling the `nextLine()` method immediately after using the `nextInt()` or `next()` methods of `Scanner`?
  A. All previous input data is erased.
  B. A line break character left after `nextInt()` or `next()` causes `nextLine()` to read a blank line.
  C. The order of input data is reversed.
  D. Numbers are automatically converted to strings when entered.

Quiz

- 在调用 `Scanner` 的 `nextLine()` 方法后立即使用 `nextInt()` 或 `next()` 方法时，可能会出现哪些常见问题?
  A. 所有之前的输入数据会被清除。
  B. `nextInt()` 或 `next()` 后留下的换行符会导致 `nextLine()` 读取一个空行。
  C. 输入数据的顺序被反转。
  D. 输入的数字会自动转换为字符串。

03

Practice

03

练习

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Practice

- 输入和输出年龄和英文姓名
  - 编写一个程序，从键盘输入用户的姓名和年龄并输出。
  - 使用 Scanner 类接收输入，然后格式化输入字符串并输出。

```
请输入年龄：20
请输入名字：小明

您的年龄是 20岁，  名字是小明。
```

练习

- 输入和输出年龄和英文姓名

  编写一个程序,从键盘输入用户的姓名和年龄并输出。

  使用 Scanner 类接收输入，然后格式化输入字符串并输出。

```
请输入年龄:20
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您的年龄是 20岁, 名字是小明。
```

Q & A

问答

THANK YOU

谢谢